Rosen — §8.4: Generating Functions

Selected Exercises

Alston

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Exercise 17

In how many ways can 25 identical donuts be distributed to four police officers so that each officer gets at least three but no more than seven donuts?

Exercise 26

(a) Show that $1/(1 - x - x^2 - x^3 - x^4 - x^5 - x^6)$ is the generating function for the number of ways that the sum n can be obtained when a die is rolled repeatedly and the order of the rolls matters.

(b) Use part (a) to find the number of ways to roll a total of 8 when a die is rolled repeatedly, and the order of the rolls matters.

Exercise 39

Use generating functions to find an explicit formula for the Fibonacci numbers.

Exercise 40

(a) Show that if n is a positive integer, then

$$\binom{2n}{n} = (-4)^n \binom{-1/2}{n}.$$

(b) Use the extended binomial theorem and part (a) to show that the coefficient of x^n in the expansion of $(1 - 4x)^{-1/2}$ is $\binom{2n}{n}$ for all nonnegative integers n.

Exercise 41

(Calculus required) Let $\{C_n\}$ be the sequence of Catalan numbers, that is, the solution to the recurrence relation $C_n = \sum_{k=0}^{n-1} C_k C_{n-k-1}$ with $C_0 = C_1 = 1$ (see Example 5 in Section 8.1).

(a) Show that if $G(x)$ is the generating function for the sequence of Catalan numbers, then $xG(x)^2 - G(x) + 1 = 0$. Conclude (using the initial conditions) that $G(x) = (1 - \sqrt{1 - 4x})/(2x)$.

(b) Use Exercise 40 to conclude that

$$G(x) = \sum_{n=0}^{\infty} \frac{1}{n+1} \binom{2n}{n} x^n,$$

so that

$$C_n = \frac{1}{n+1} \binom{2n}{n}.$$

(c) Show that $C_n \geq 2^{n-1}$ for all positive integers n.

Exercise 43

Use generating functions to prove Vandermonde's identity:

$$\binom{m+n}{r} = \sum_{k=0}^{r} \binom{m}{r-k}\binom{n}{k},$$

whenever m, n, and r are nonnegative integers with r not exceeding either m or n. [*Hint*: Look at the coefficient of x^r in both sides of $(1+x)^{m+n} = (1+x)^m(1+x)^n$.]

Exercise 49

A coding system encodes messages using strings of octal (base 8) digits. A codeword is considered valid if and only if it contains an even number of 7s.

(a) Find a linear nonhomogeneous recurrence relation for the number of valid codewords of length n. What are the initial conditions?

(b) Solve this recurrence relation using Theorem 6 in Section 8.2.

(c) Solve this recurrence relation using generating functions.

Exercise 56

Recall: $p(n)$, the number of partitions of n, is the coefficient of x^n in the formal power series expansion of $1/((1-x)(1-x^2)(1-x^3)\cdots)$.

(Requires calculus) Use the generating function of $p(n)$ to show that $p(n) \leq e^{C\sqrt{n}}$ for some constant C. [Hardy and Ramanujan showed that $p(n) \sim e^{\pi\sqrt{2/3}\sqrt{n}}/(4\sqrt{3}n)$, which means that the ratio of $p(n)$ and the right-hand side approaches 1 as n approaches infinity.]

Exercise 57

Recall: the probability generating function for a random variable X such that $X(s)$ is a nonnegative integer for all $s \in S$ is $G_X(x) = \sum_{k=0}^{\infty} p(X(s)=k)\, x^k$.

(Requires calculus) Show that if G_X is the probability generating function for such a random variable X, then

(a) $G_X(1) = 1$.

(b) $E(X) = G'_X(1)$.

(c) $V(X) = G''_X(1) + G'_X(1) - G'_X(1)^2$.